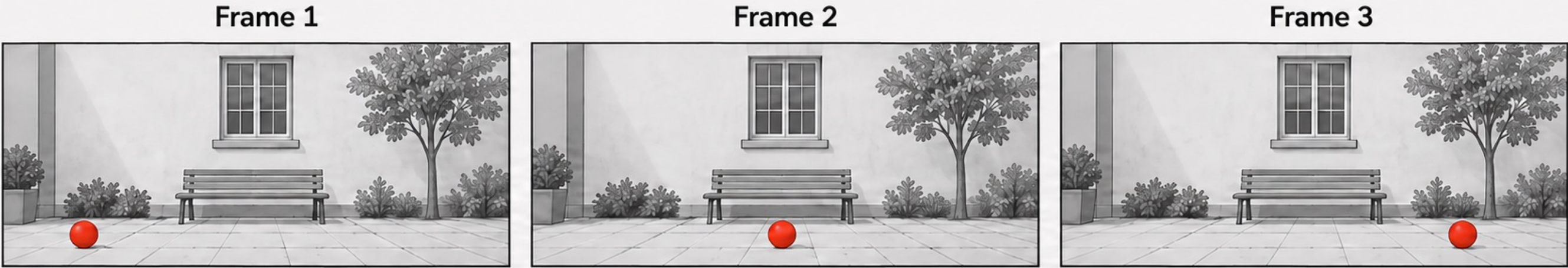


LLaVA-OneVision-2

From codec-aligned extremely long video understanding to unifying visual understanding across time and space

HEVC inter-frame prediction (conceptual)

Repeated visual content



The background repeats while the ball moves

Prediction + correction



Under a fixed token budget, how can we improve temporal coverage and evidence density?

01

Background and problem

Model evolution, 2024–2026

2024	2025	2026 · 02	2026 · 05
■	■	■	■
LLaVA-OneVision	OV1.5	OneVision-Encoder	OV2
Images, multi-image, short video unified task transfer	85M mid-training + 46M instruction native-resolution image foundation	Codec patchification unified image–video encoder	Four training stages 15-minute video + spatial/tracking

EVOLUTION AXIS

Visual-token organisation → data scale and quality → progressive video-length training → spatial and object-trajectory supervision.

Coverage versus token budget

Uniform frame sampling



6
frames

54
tokens

100%
patches retained

Static backgrounds repeat; fast events can fall between frames.

Codec-aligned sampling



18
frames

54
tokens

3x
time span

Retain I-frame global context (red); encode only new evidence in P-frames.

Lineups

01 · VISUAL REPRESENTATION

OneVision-Encoder

Codec motion and residual signals select patches; a shared 3D RoPE encodes irregular spatiotemporal tokens.

02 · MULTIMODAL FOUNDATION

LLaVA-OneVision-1.5

Image and OCR capability from 85M concept-balanced mid-training samples, 46M instruction samples and offline packing.

03 · PROGRESSIVE VIDEO TRAINING

LLaVA-OneVision-2

Reuses OV1.5 and expands video length from 30 seconds to 10–15 minutes through a four-stage training process.

TAKEAWAY

OV2 gains come from representation sparsification, data organized by training stage and spatial/tracking supervision.

02

OneVision-Encoder

H.264 / AVC and H.265 / HEVC

Coding structure	H.264 / AVC	H.265 / HEVC
Block hierarchy	16 × 16 macroblocks, with smaller partitions	Coding tree units up to 64 × 64, recursively split
Parameter sets	SPS, PPS	VPS, SPS, PPS

SHARED PREDICTIVE CODING

Prediction, residual transform / quantization, and entropy coding

I · INTRA

Prediction within the current picture

P · INTER

Prediction using reference list 0

B · INTER

Prediction using reference lists 0 and/or 1

Reconstruction: prediction + decoded residual

Clipping and in-loop filtering follow. MP4 is a container, separate from the video coding standard.

Video metadata and bitstream headers

FFPROBE SNAPSHOT · birds_header.json

```
{
  "codec_name": "h264",
  "profile": "High",
  "width": 3840,
  "height": 1620,
  "pix_fmt": "yuv420p",
  "avg_frame_rate": "25/1",
  "time_base": "1/12800",
  "nb_frames": "765",
  "extradata_size": 52
}

ffprobe -v error -show_streams -show_format \
  -of json birds.mp4
```

INFORMATION LAYERS

Layer	Typical information
Container	Tracks, timing, sample sizes
Parameter sets	SPS/PPS (+ HEVC VPS): profile, size, coding tools
Slice headers	Type, picture order, reference configuration
Coded blocks	Motion-related syntax and residual coefficients

Block-level motion and residuals require decoder access.

Decoder interfaces for motion and residuals

C HEADER · OPENHEVC WRAPPER (ABRIDGED)

```
typedef struct OpenHevc_Frame_cpy {
    void *pvY, *pvU, *pvV;
    void *pvMV;
    void *pvYR, *pvUR, *pvVR;
    OpenHevc_FrameInfo frameInfo;
} OpenHevc_Frame_cpy;
```

FFMPEG · AVMotionVector

Block size, source / destination positions,
motion_x / motion_y, motion_scale

Decoded pixels provide the visual input. Motion and residuals guide token allocation.

DECODER OUTPUT BUFFERS

Field group	Meaning
pvY / pvU / pvV	Reconstructed YUV pixels
pvMV	Motion-information buffer
pvYR / pvUR / pvVR	Decoded residual planes
frameInfo	Size, bit depth, strides, timestamp

MV export depends on decoder support.
Residual planes here use the modified
OpenHEVC wrapper.

Codec patchification

1 · HEVC BITSTREAM

Read the codec

I-frame, P-frame motion vectors, P-frame residuals.

64

frames per clip

2 · PATCH SALIENCY

Score each patch

Motion magnitude per patch plus luma residual energy.

32

GOP length

3 · GLOBAL TOP-K

Rank and keep

Keep all I-frame patches; rank P-frame patches by saliency across the clip.

2 + 62

I-frames + P-frames

4 · SPARSE RGB TOKENS

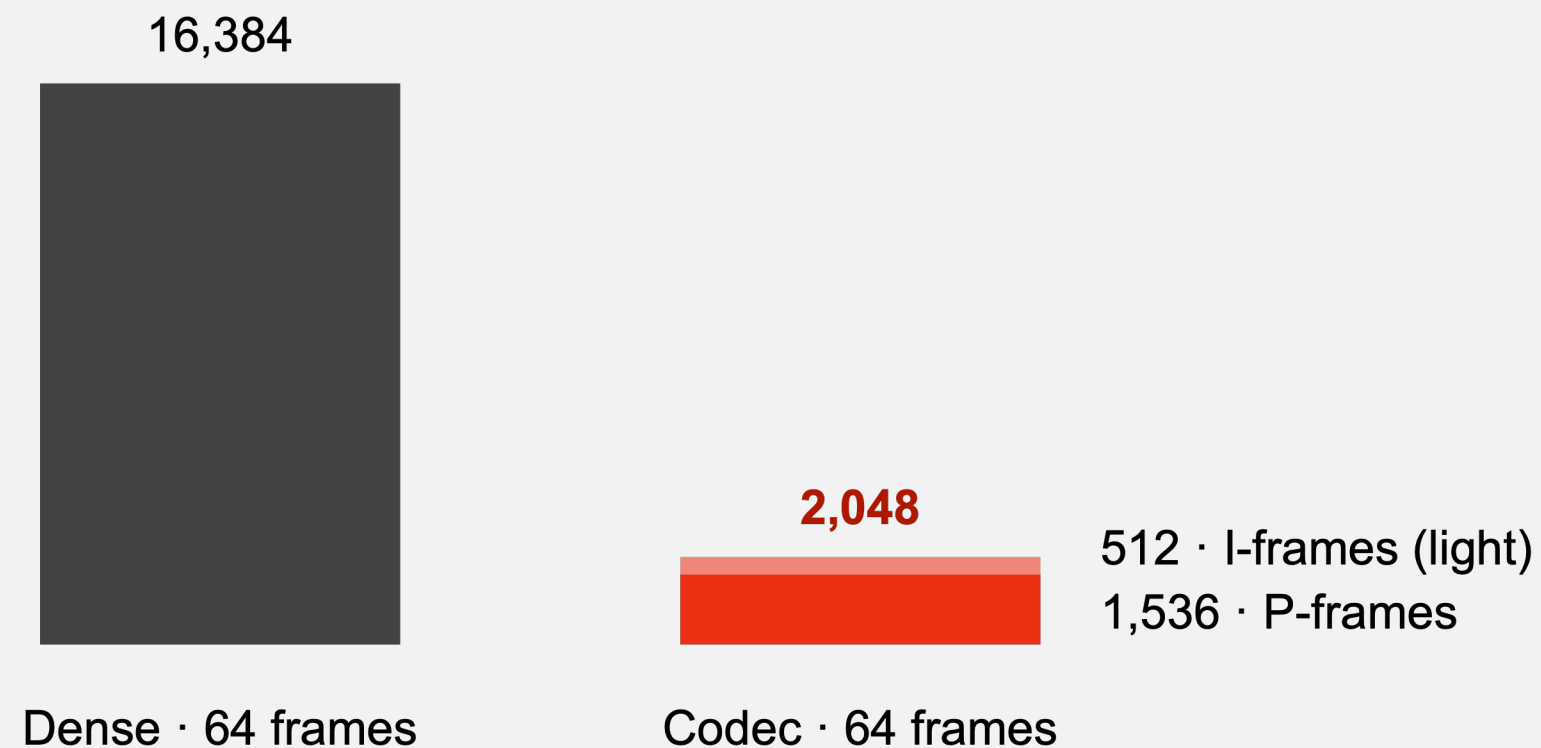
Encode

RGB patch with its original (t, x, y) into the shared ViT.

2,048

clip-level token budget

Token accounting at 64 frames



87.5%

token reduction

ALLOCATION RULES

All 256 patches from each of the two I-frames are retained

The remaining 1,536 tokens compete globally across 62 P-frames

No fixed per-frame quota; dynamic segments receive more tokens

Sparse coordinates preserve position through 3D RoPE

A shared encoder across input formats

IMAGE

Complete spatial grid

One frame, all patches.

$(0, x, y)$

CHUNK / FRAMES

One frame per temporal chunk

Uniformly sampled frames, chunk index as time.

(chunk, x, y)

CODEC

Irregular sparse patches

I- and P-frame patches retain their source temporal indices.

(t, x, y)

ViT-L/14

24

4:6:6

14 × 14

0

OneVision-Encoder backbone

layers · hidden 1024 · 16 heads

3D RoPE split (t : x : y)

patch · attentive pooling

modality-specific parameters

3D RoPE for irregular tokens

CODEC

Source temporal offsets

$$\Delta p = (t_i - t_j, x_i - x_j, y_i - y_j)$$

Retains relative temporal positions across I- and P-frames.

CHUNK-WISE

Chunk indices as time

$$\Delta p = (c_i - c_j, x_i - x_j, y_i - y_j)$$

Represents temporal order at chunk granularity.

SINGLE IMAGE

Pure spatial offsets

$$\Delta p = (0, x_i - x_j, y_i - y_j)$$

Reduces to 2D relative position.

KEY EFFECT

Sparse selection preserves the positional indices of retained patches. Token order alone does not define temporal distance.

Pretraining objective

OFFLINE CLUSTERING

MetaCLIP-H/14 teacher

Images: 2.0M object centres
Videos: motion-level centres
Top-10 positive labels per sample

SHARED VISUAL ENCODING

Image · OCR · Video

One OneVision-Encoder
Attentive pooling to a clip or image embedding

MULTI-LABEL LOSS

Cluster discrimination

$$\mathcal{L} = \sum_m \mathbb{E} \log(1 + \exp(-y \cdot \sigma))$$

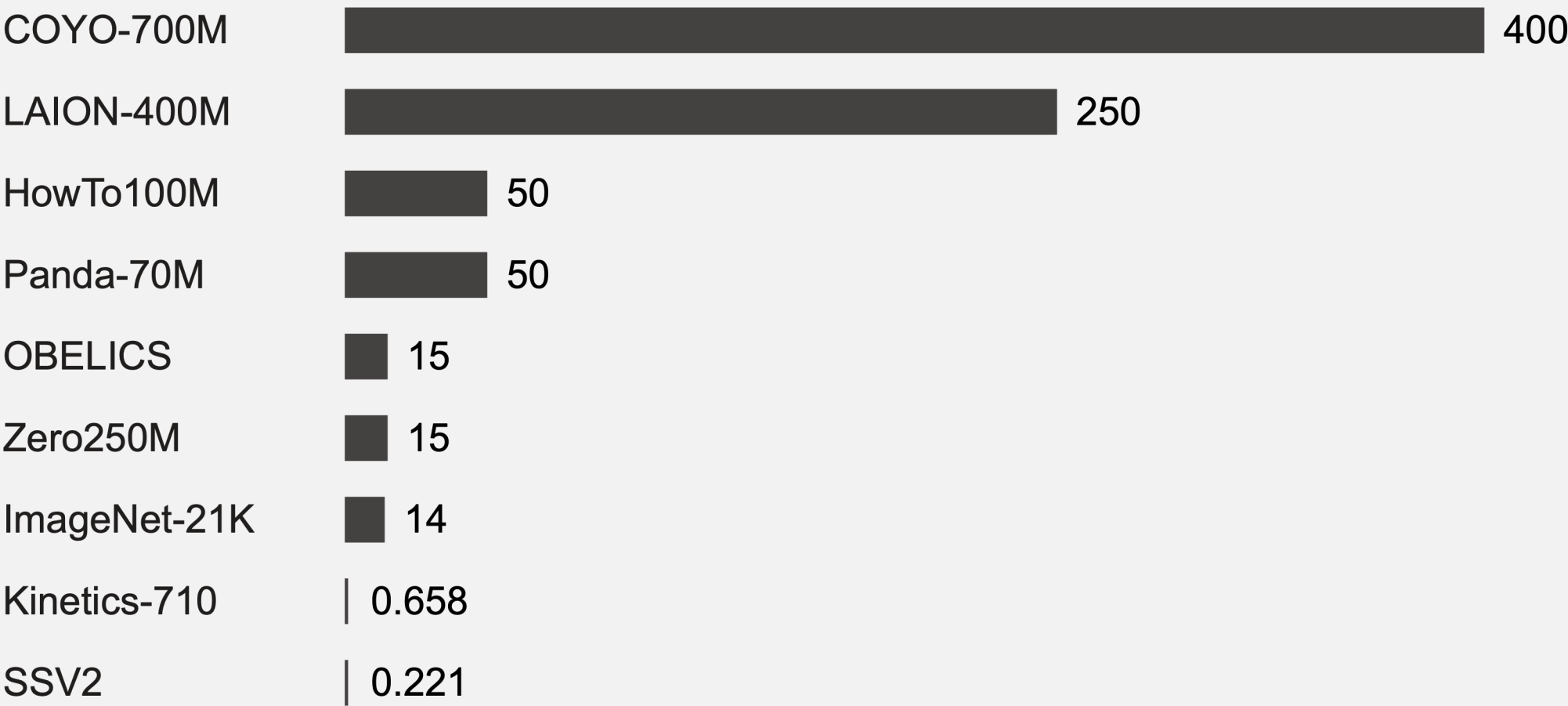
Positive and negative centres sampled at object and video granularity; strengthens within-class structure and cross-modal consistency.

DESIGN IMPLICATION

Visual features acquire structured semantics before OV-Encoder-Lang or the LMM aligns them with language.

Pretraining data

Listed source counts (millions)



694M

image entries — objects and OCR

100.9M

video entries — web and action video

≈795M

sum of listed source counts

Two-stage pretraining

STAGE 1 · IMAGE ONLY

Object semantics

13B

training specs
224² · AdamW · LR 1e−3 · WD 0.2 · 2M classes
· Top-10 positives

STAGE 2 · IMAGE + OCR + VIDEO

Dynamics and text

4B

training specs
image 448² / video 224² · LR 1e−4 · video :
image = 1 : 1

STAGE 2 VIDEO SAMPLING MIXTURE



Codec 50%

Frame 37.5%

Tile 12.5%

128

A800 GPUs · 16 nodes × 8

17B

total training exposures

—

batch, warmup, schedule, precision, wall-
clock: undisclosed

03

OV1.5 foundation

OV1.5 architecture

VISION ENCODER

RICE-ViT

450M images / 2.4B regions ·
region-aware attention · 2D
RoPE · native resolution

PROJECTOR

2×2 merge + MLP

Concatenate four neighbouring
patches (4× token merge); two-
layer MLP into LLM space

LANGUAGE MODEL

Qwen3 · 4B / 8B

Reasoning and language
generation

WHY OV1.5 MATTERS

Large-scale image mid-training and instruction tuning establish OCR, document, chart and general visual-QA capability before the OV2 video training process.

OV1.5 data recipe

MID-TRAINING · 85M IMAGE–TEXT PAIRS

Concept-balancing pipeline

65M English + 20M Chinese · 8 source pools · MetaCLIP 500K concepts · Top-K assignment · inverse-frequency sampling · bilingual captioning

A 2M concept-balanced subset beats random sampling on 25 of 27 benchmarks.

85M

mid-training pairs

≈46M

instruction samples

INSTRUCTION · 22M → MERGED 46M

Eight task categories

Caption · Chart & table · Code & math · Domain-specific · General VQA · Grounding & counting · OCR · Science

The 22M in-house set is deduplicated and merged with FineVision into 46M samples. The paper says “7 categories” but lists 8.

25 / 27

benchmarks won by concept balancing

OV1.5 training and systems

STAGE 1

Language–image alignment

558K

samples · projector only*

≤11×

packing compression · hash
buckets → multithreaded
batching → uniform-length
records

STAGE 1.5

Knowledge learning

85M

samples · full parameters · 8K
context

STAGE 2

Instruction tuning

46M

samples · full parameters

128

A800 GPUs

3.7 d

8B mid-training wall-clock

<\$16K

claimed budget (scope not
itemised)

* Freeze state inferred from “pretrain projection layer”; not enumerated · An et al., 2025, Sec. 4

04

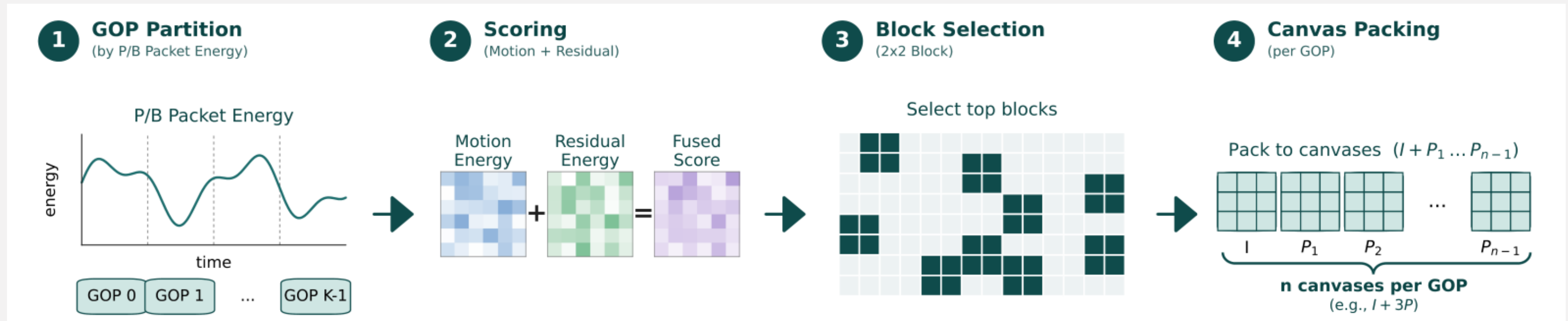
OV2 training stages

From OV1.5 to OV2

	OV1.5	OV2
Input	Native-resolution image	Image + frames + codec stream
Temporal range	Single image / short video	30 s → 3 min → 10–15 min
Representation	RICE-ViT · 2D RoPE	OneVision-Encoder · 3D RoPE
Supervision	Image–text + instructions	Dense captions + spatial + tracking

OV2 is not mainly a larger language model; it reorganises visual input and the training training process.

OV2 codec-stream tokenization



Adaptive temporal groups

P/B bit cost gives changing intervals finer temporal grouping.

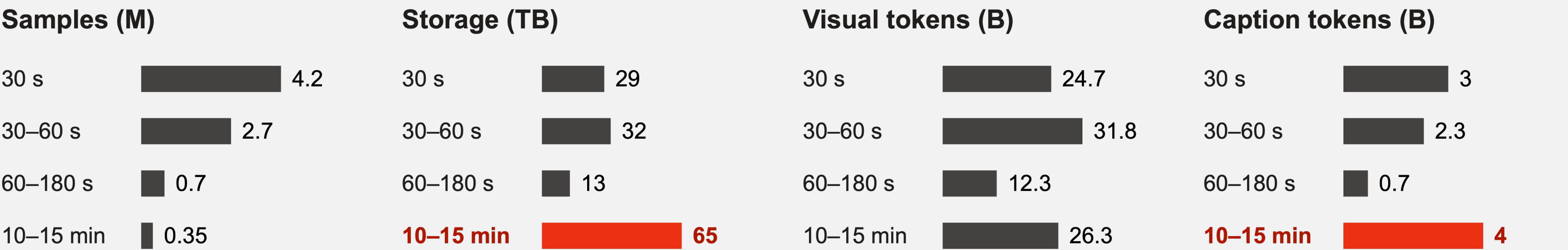
Coverage across time

Select 2×2 blocks and distribute P-canvases across the group.

Shared visual encoding

Source coordinates and GOP visibility groups connect the packed tokens.

Video-caption corpus by duration



≈8M
video samples

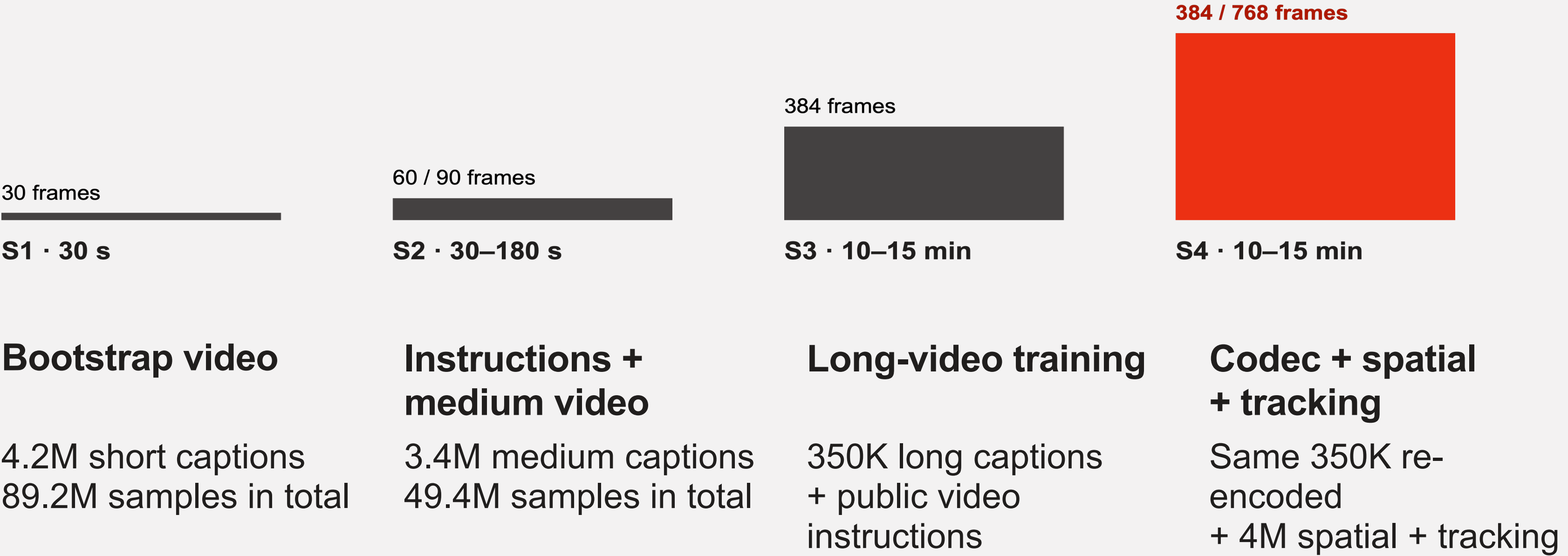
≈139 TB
project-page storage (HF
card: 66.6 TB)

95.1B*
visual tokens · chart sum*

10.0B*
caption tokens · chart sum*

* Project-page chart sums; arXiv Fig. 7 reports 104.1B total tokens

Four training stages



Stages 1–2: short video and instructions

STAGE 1 · BOOTSTRAP VIDEO

Basic temporal modelling

85M OV1.5 mid-training
4.2M 30-second captions
30 frames · 392^2 · nominal 5,880 visual tokens
per clip

Role: establish temporal modelling without
sacrificing image capability.

STAGE 2 · INSTRUCTION + MEDIUM VIDEO

Instruction following at medium length

22M OV1.5 instruct + 24M FineVision
3.4M video captions
60 frames (30–60 s) · 90 frames (60–180 s) ·
 392^2

Role: jointly train multimodal instruction following
and medium-length description.

Stages 3–4: long video, codec, spatial, tracking

STAGE 3 · LONG-VIDEO UNDERSTANDING

350K captions · 384 frames

22M OV1.5 instructions + 24M FineVision
LLaVA-Video-178K: 1.6M instructions
Selected VideoChat-Flash instructions

STAGE 4 · CODEC / SPATIAL / TRACKING

4M spatial · 384 / 768 frames

Re-encode the same 350K long-video clips
Add spatial QA and point-tracking supervision
Molmo2-VideoTrack and VideoPoint

Stage 4 revisits the same 350K clips with denser codec inputs and their existing captions.

Data reuse across training stages

REUSED · OV1.5 FOUNDATION

Image-language foundation

OV1.5 mid-training · 85M
OV1.5 instruct · 22M
FineVision · 24M

OV2 TRAINING STAGES

S1 → S2 → S3 → S4

Stage 1 mixes image mid-training with short videos. Later stages mix image instructions with longer videos and spatial tasks.

NEW · VIDEO, SPATIAL, TRAJECTORY

Temporal and spatial supervision

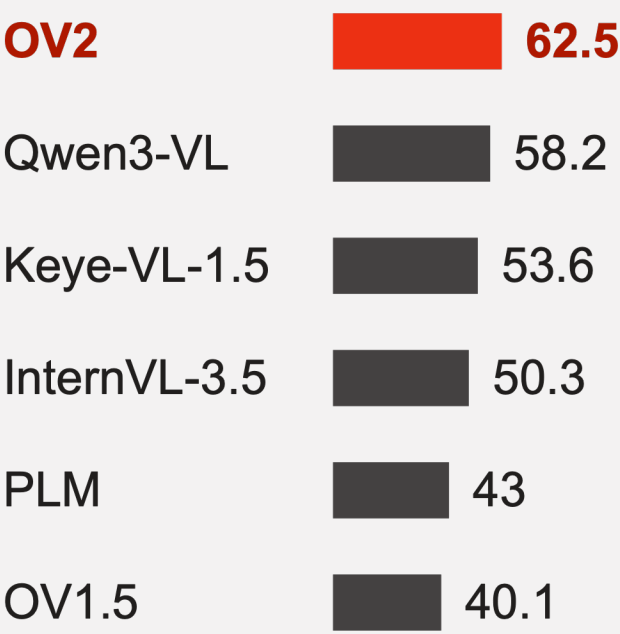
Video captions · ≈8M
Spatial · 4M
Public video instruct · ≥1.6M + unknown
Tracking / pointing · unknown

05

Evaluation

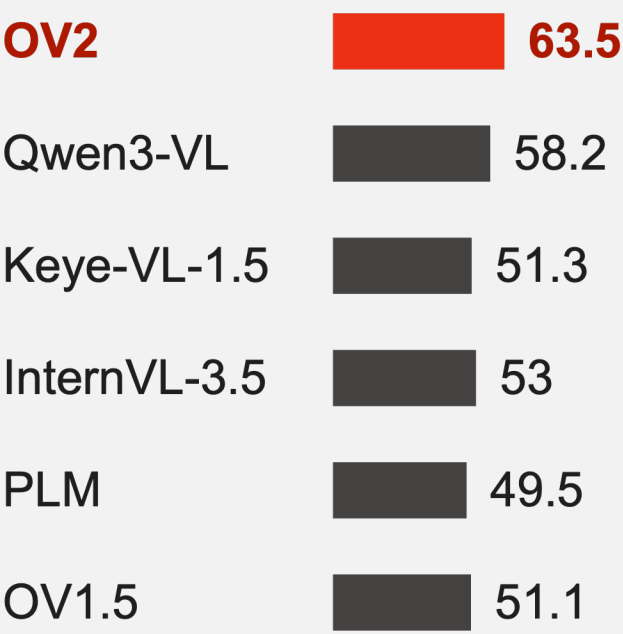
Results by task family

Video · 18 tasks



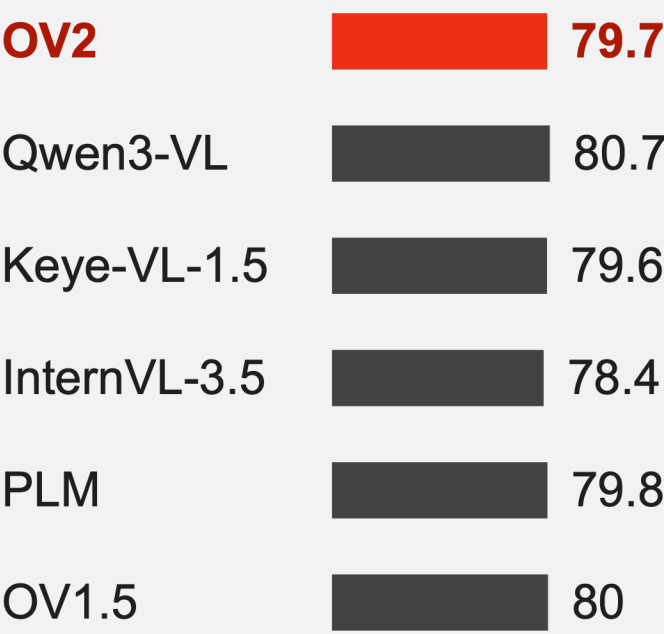
+22.4
video vs. OV1.5

Spatial · 11 tasks



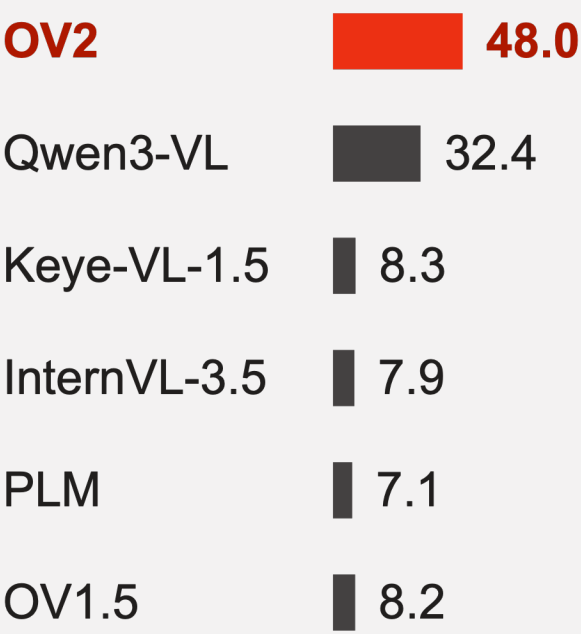
+12.4
spatial vs. OV1.5

Image · 11 tasks



-0.3
image vs. OV1.5

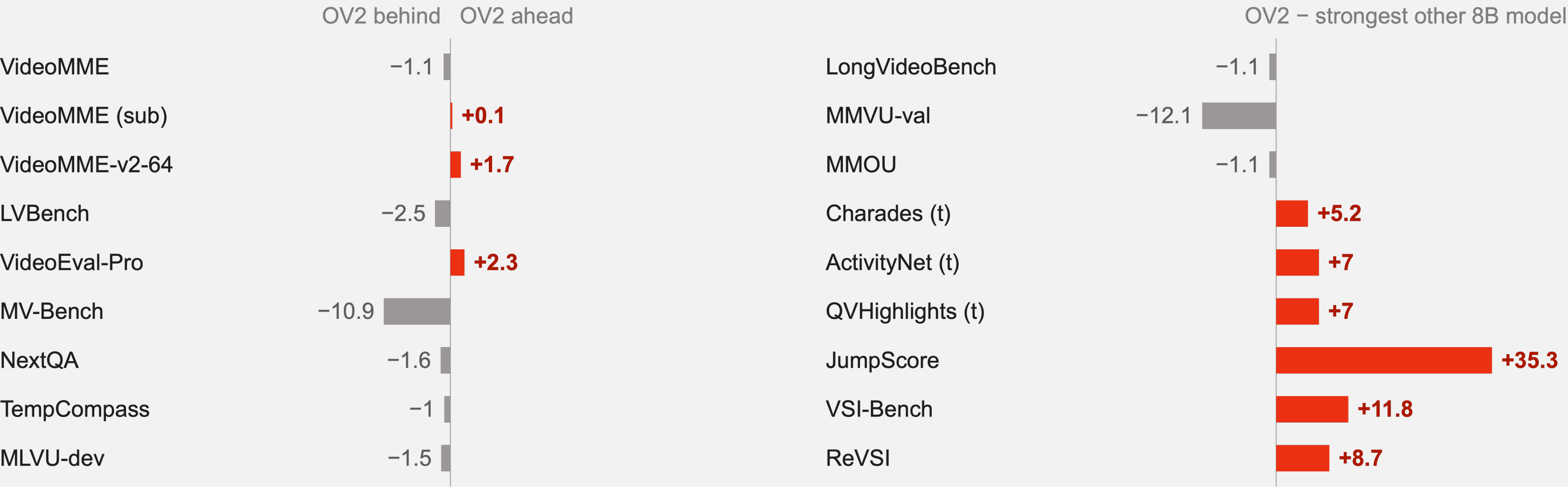
Tracking · 4 datasets (J&F)



+39.8
tracking J&F vs. OV1.5*

* arXiv Table 1: mean J&F over 4 datasets

Video benchmarks: OV2 margin on 18 tasks



Spatial benchmarks: OV2 margin on 11 tasks

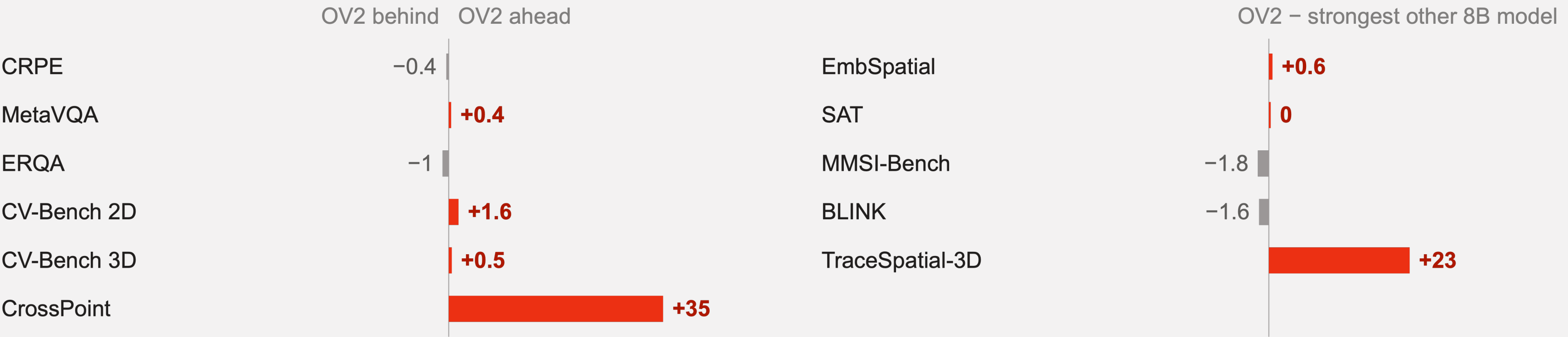
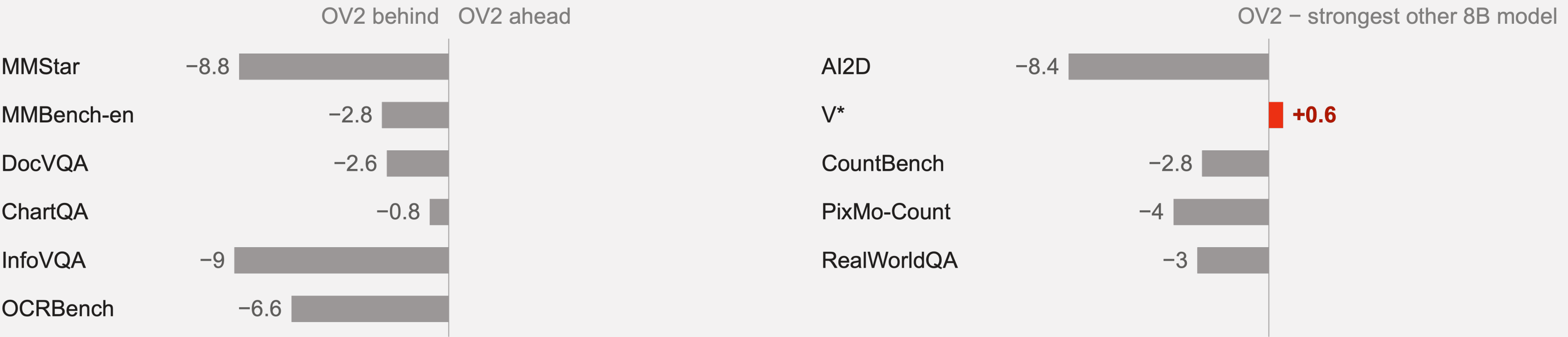
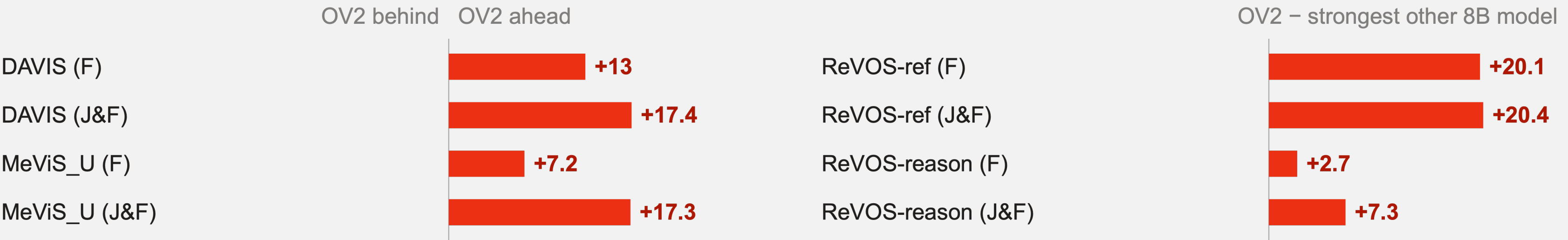


Image benchmarks: OV2 margin on 11 tasks



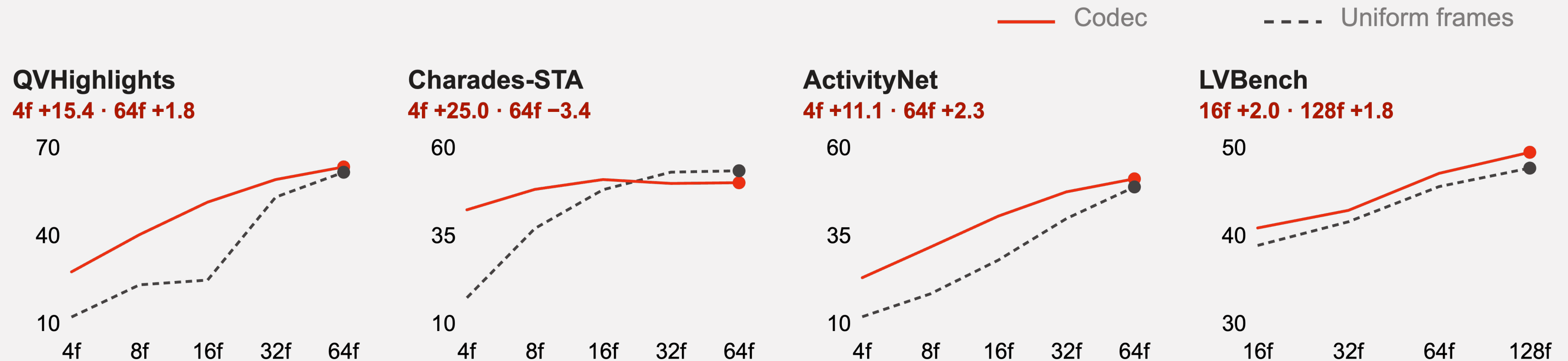
The image average stays close to OV1.5: 79.7 vs. 80.0. Individual tasks show gains and regressions.

Tracking and segmentation: OV2 margin on 8 metrics



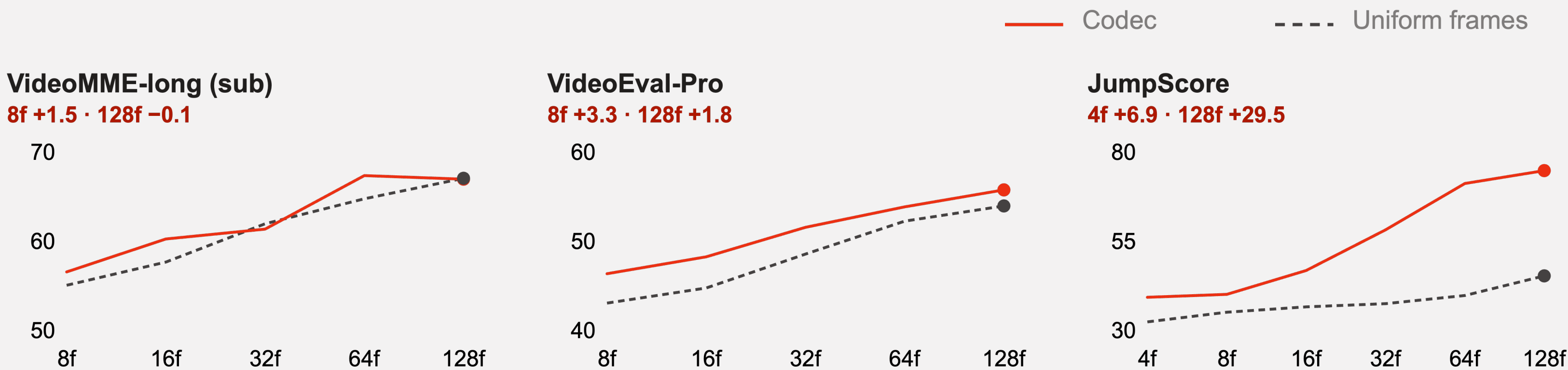
Mean J&F: 48.0 vs. 32.4 for Qwen3-VL-8B. HOTA results are mixed across datasets.

Codec versus frames: grounding and long video



Grounding gains are largest at small budgets. At 64 frames the gap narrows, and Charades-STA favors uniform sampling.

Codec versus frames: QA and repeated motion

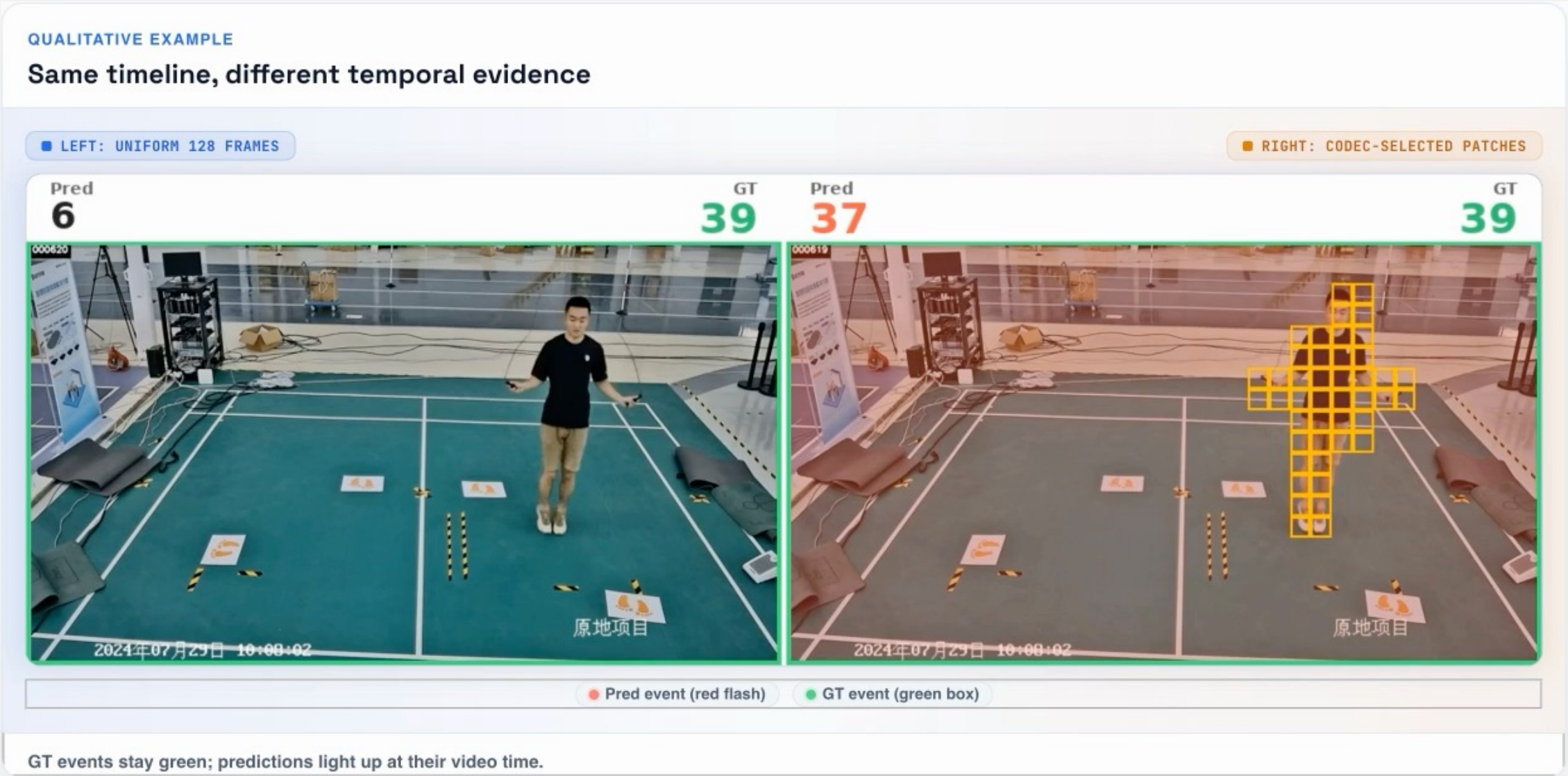


-0.1
VideoMME-long @128f

+1.8
VideoEval-Pro @128f

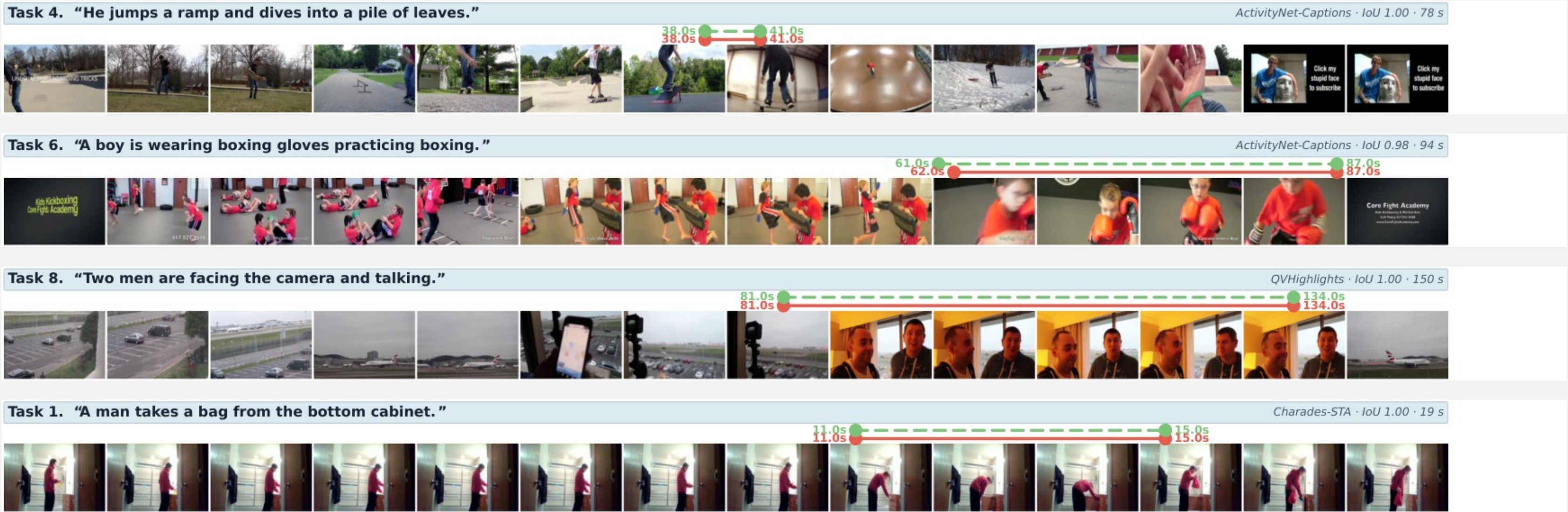
+29.5
JumpScore @128f

JumpScore: cycle-start localization



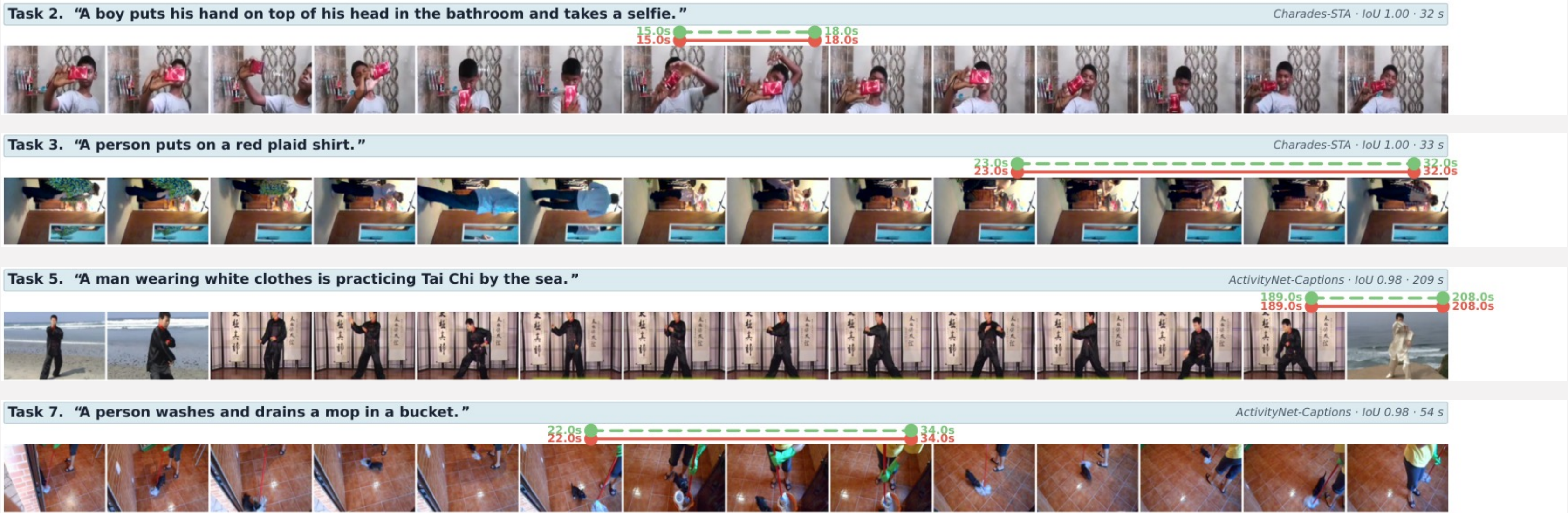
Temporal grounding cases 1/2

Prediction: solid red Ground truth: dashed green



Temporal grounding cases 2/2

Prediction: solid red Ground truth: dashed green

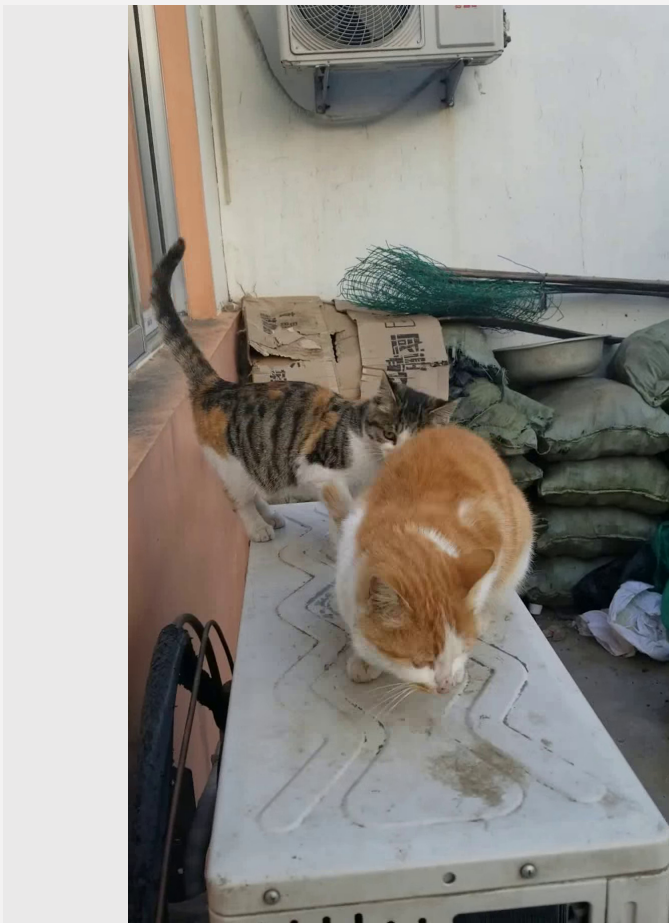


Referring video segmentation

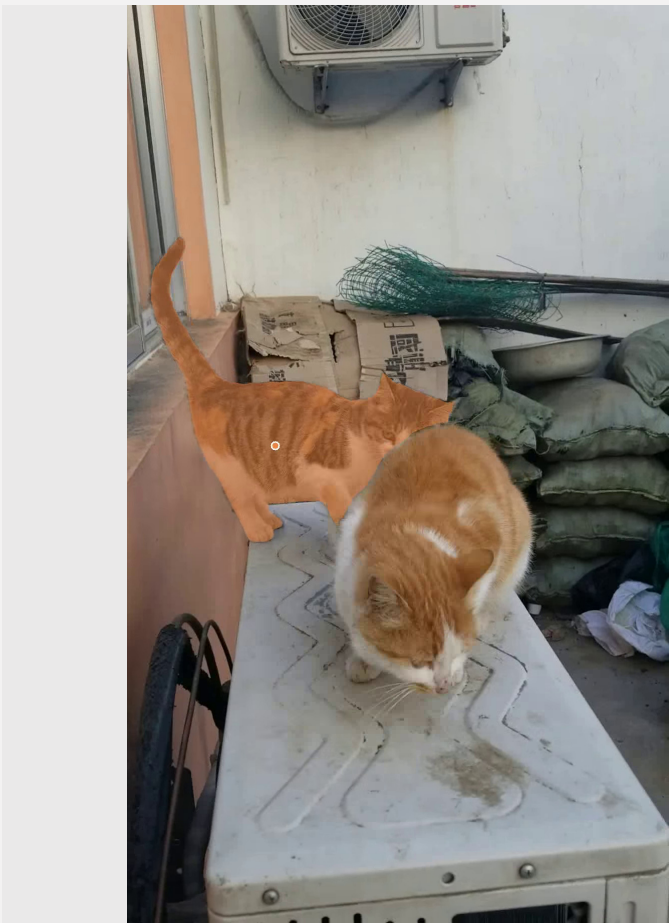
R-VOS · CASE 1

“Track the animal moving forward.”

INPUT



MASK



R-VOS · CASE 2

“Track the person whose appearance deviates the most from the norm.”

INPUT



MASK

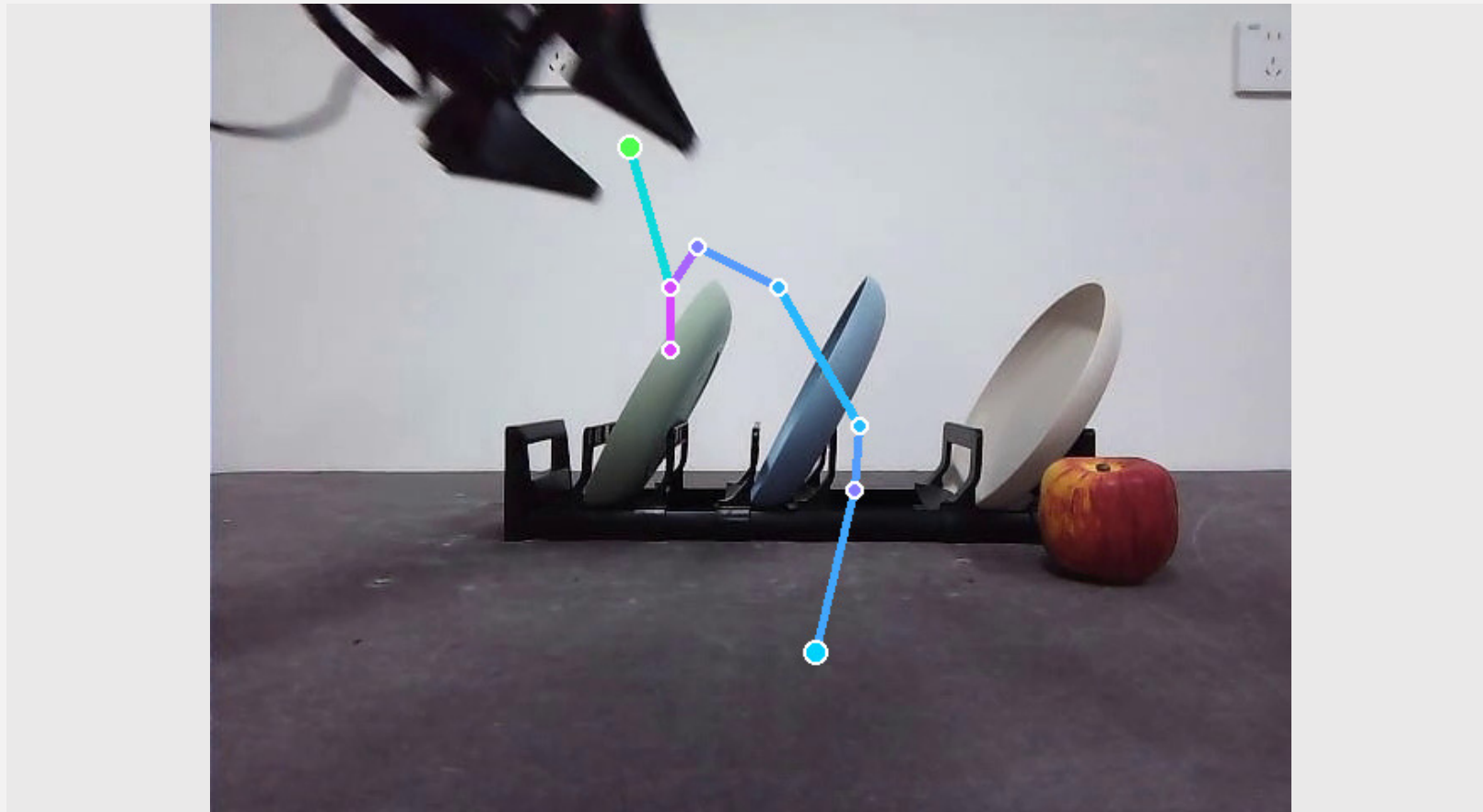


Video manipulation

ROBOT TASK

9 predicted (x, y, z) waypoints at $t = 0$ s

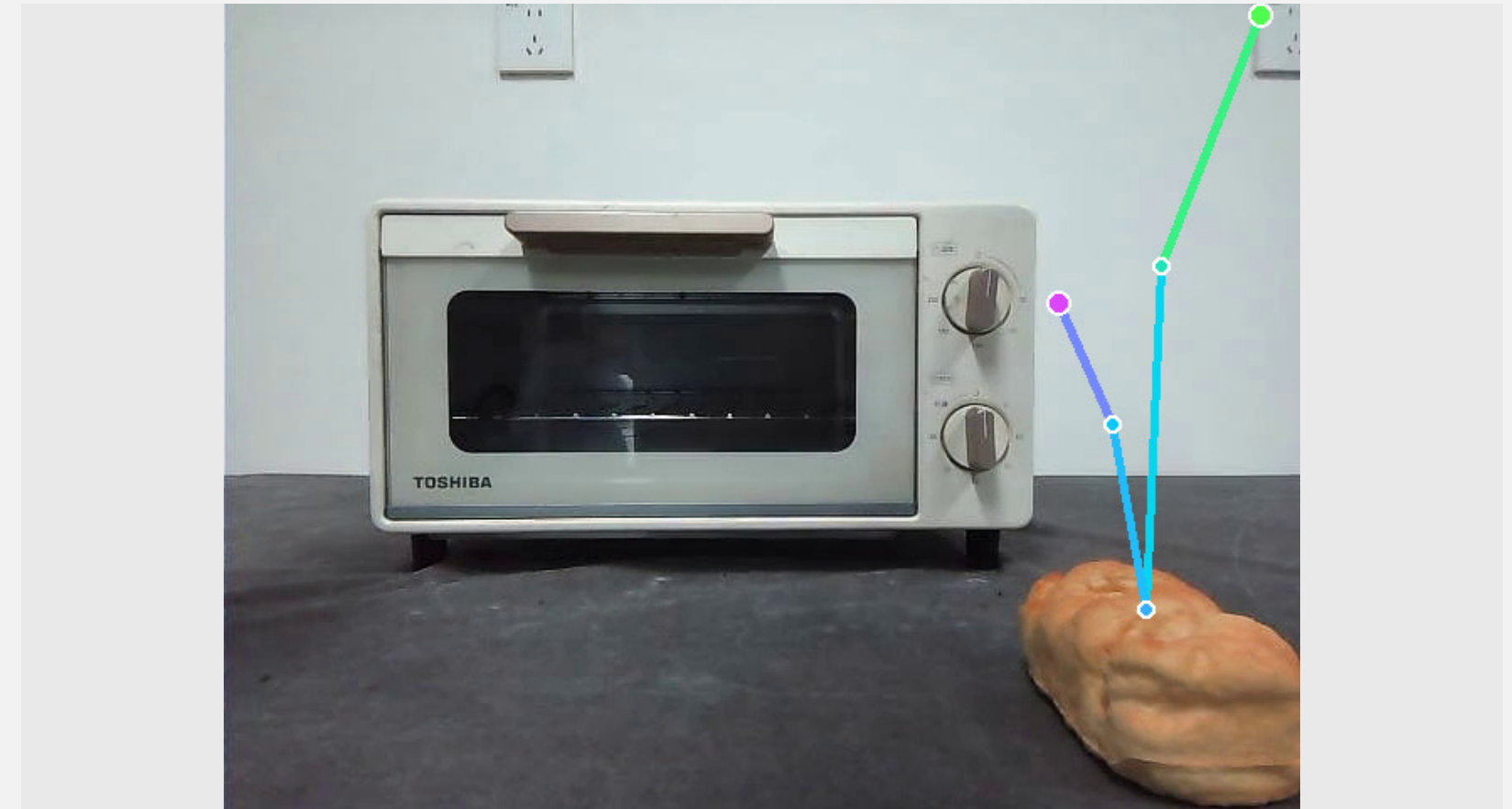
“Put the apple on the green plate placed on the table.”



ROBOT TASK

5 predicted (x, y, z) waypoints at $t = 0$ s

“Put the bread into the oven.”



Spatial grounding: object references

2D POINTING

pixel-coordinate point

“Please point to the top piece of paper on the white table.”



3D TRAJECTORY

pick-and-place trajectory

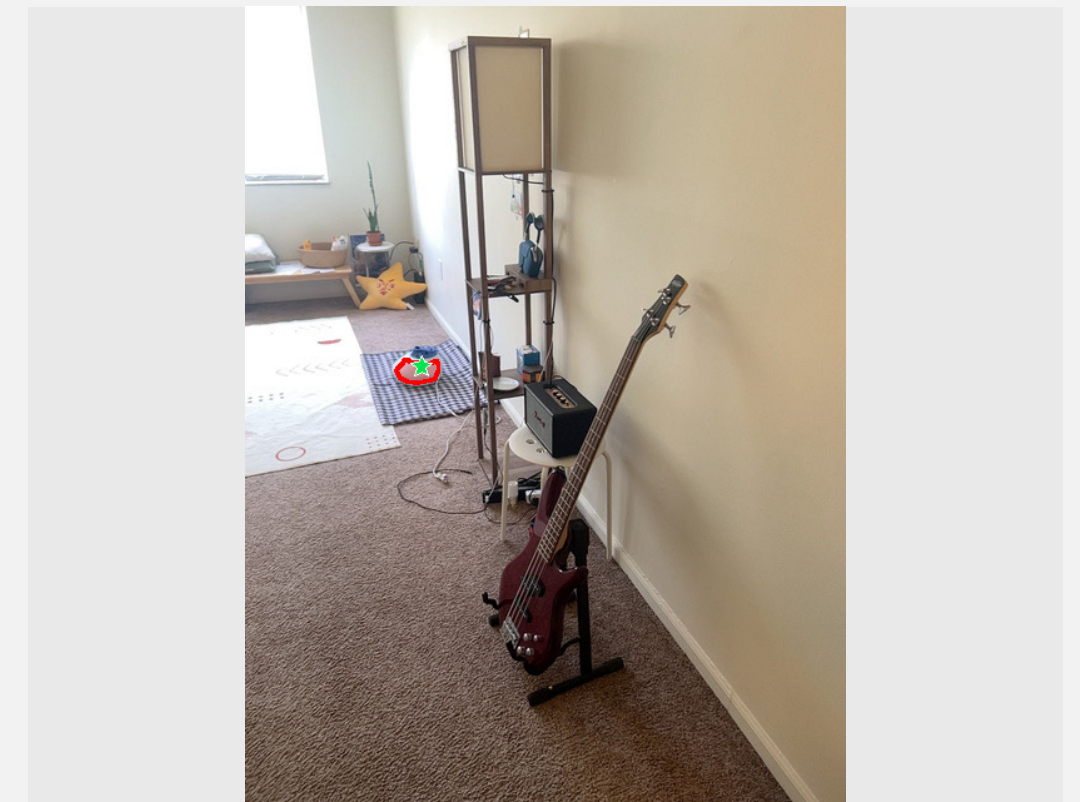
“Pick up the brown small bottle and move it to the left of the white mouse.”



2D POINTING

pixel-coordinate point

“Point out the white object second closest to the wooden shelf.”



Spatial grounding: free space

2D POINTING

pixel-coordinate point

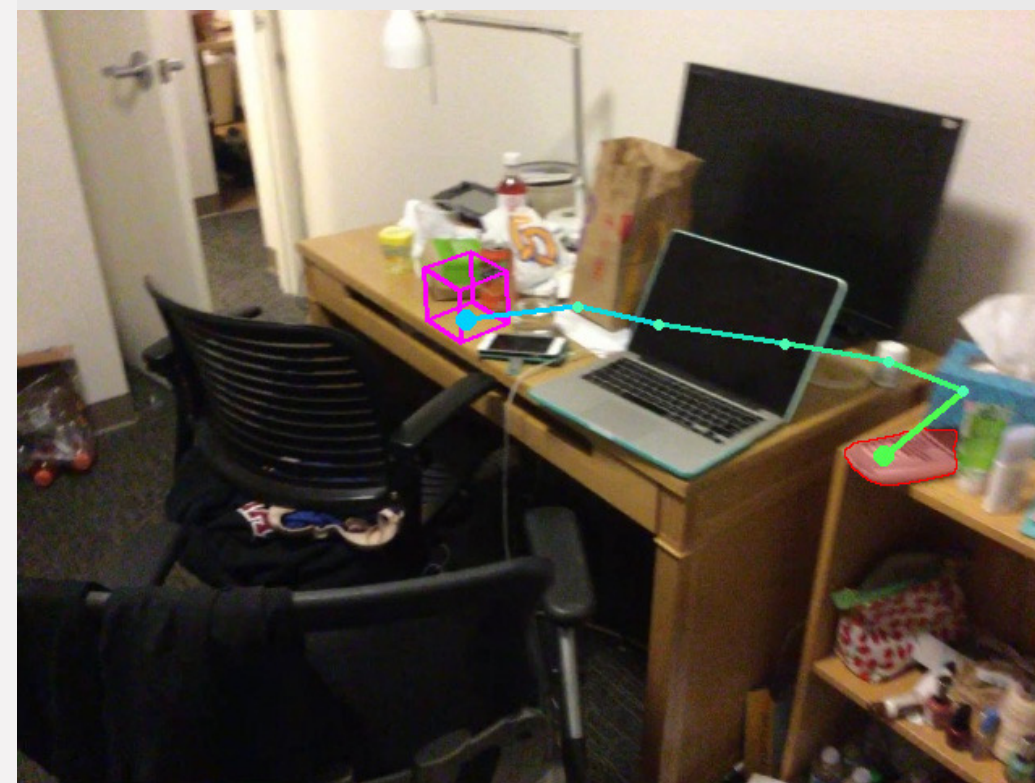
“Point out the free space between the cat tree and litter box.”



3D TRAJECTORY

pick-and-place trajectory

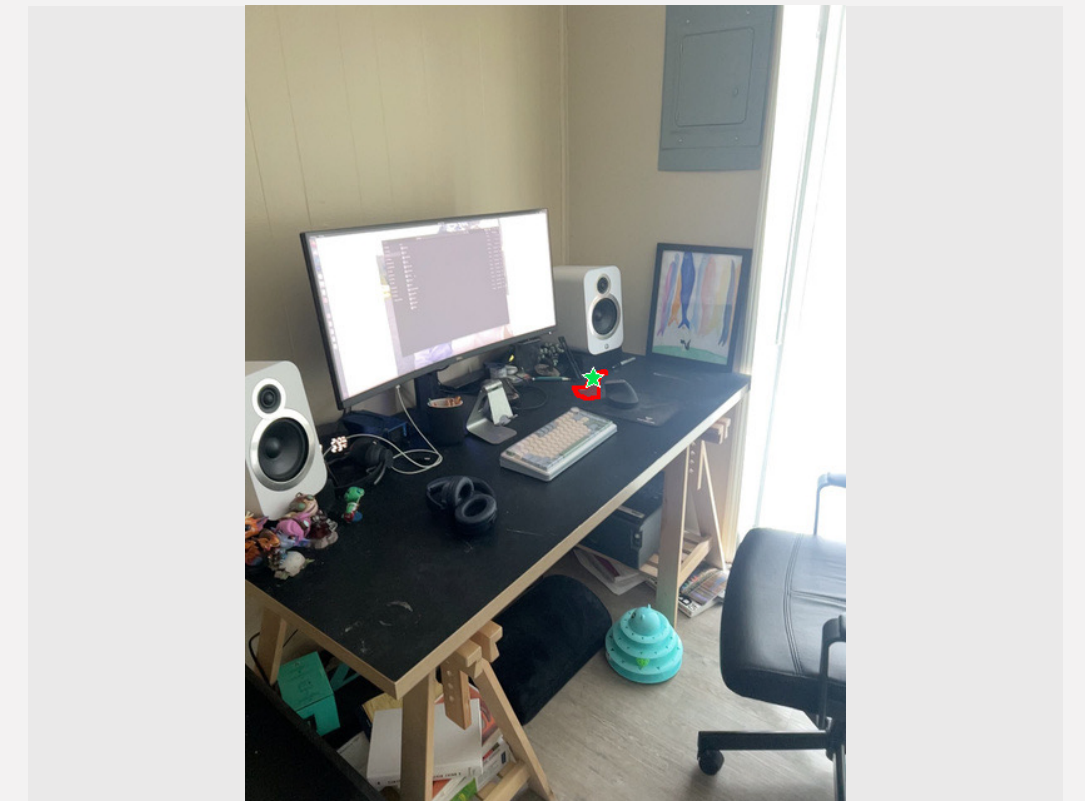
“Move the calculator to the left of the phone on the left table.”



2D POINTING

pixel-coordinate point

“Point out free space between the speaker and the mouse.”



CONCLUSIONS

What should a video token preserve?

VISUAL REPRESENTATION

Context and change

Codec signals guide where we allocate visual evidence.
Shared positions and attention groups preserve its temporal structure.

PROGRESSIVE TRAINING

Learning across time

OV1.5 provides the image-language foundation.
Progressive video training and spatial supervision extend that foundation.

TASK-DEPENDENT EVIDENCE

Where allocation helps

Grounding benefits most at small observation budgets.
Dense frames remain useful for detail-sensitive queries and some larger budgets.

DISCUSSION

How should the task guide the visual evidence we keep?